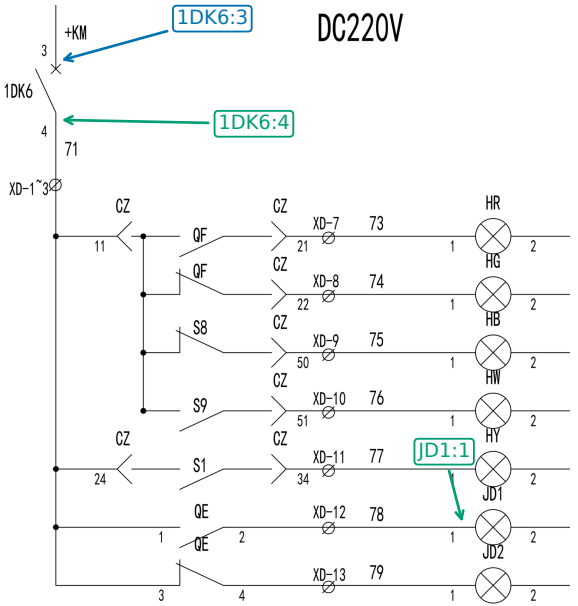
(a) Original DWG circuit window



(b) Engineering question

Using 1DK6:3 as the reference, where should continuity be checked to distinguish the observable fault patterns with the fewest continuity tests? Choose from 1DK6:4, JD1:1, CZ:11, CZ:24, JD2:1 and QE:1. Exactly one of 1DK6 contact 3–4, QE contact 1–2 and QE contact 3–4 is stuck open; the others are closed and wires intact. The circuit is isolated. Treat unlisted contacts as open.

Answer: 2 tests: 1DK6:4 and JD1:1.

(c) Checkable fault signatures

Stuck-open contact	1DK6:4	JD1:1
1DK6 3–4	0	0
QE 1–2	1	0
QE 3–4	1	1

1 = connected to 1DK6:3; 0 = disconnected.

Three distinct signatures; one binary test cannot distinguish three possibilities. Two tests suffice.

Why topology matters: common and branch-specific contacts produce different fault signatures.

Visual grounding → explicit fault conditions → test selection → verifiable evidence

Reference example, not a model prediction. Isolated continuity test; the DC220V label does not imply live operation.